

(A405516) COMPILER DESIGN LABORATORY

B. Tech (CSE) III Year VI Semester

$\frac{L}{0}$	$\frac{T}{0}$	$\frac{P}{2}$	$\frac{C}{1}$
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1. Implementation of symbol table.
2. Develop a lexical analyzer to recognize a few patterns inc (ex. Identifiers, constants, comments, operators etc.)
3. Implementation of lexical analyzer using lex tool.
4. Generate yacc specification for a few syntactic categories.
 - a) Program to recognize a valid arithmetic expression that uses operator +, -, * and /.
 - b) Program to recognize a valid variable which starts with a letter followed by any number of letter or digits.
 - c) Implementation of calculator using lex and yacc.
5. Convert the bnf rules into yacc form and write code to generate abstract syntax tree.
6. Implement type checking.
7. Implement any one storage allocation strategies (heap, stack, static)
8. Write a lex program to count the number of words and number of lines in each file or program.
9. Write a 'C' program to implement lexical analyzer using c program.
10. write recursive descent parser for the grammar.
E->E+T E->T T->T*F T->F F->(E)/id.
11. write recursive descent parser for the grammar.
S->(L) S->a L->L, S L->S
12. Write a C program to calculate the first function for the grammar.
E->E+T E->T T->T*F T->F F->(E)/id
13. Write a YACC program to implement a top-down parser for the given grammar.
14. Write a YACC program to evaluate algebraic expression.

TEXTBOOK:

1. Compilers: Principles, Techniques and Tools, Second Edition, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman.

COURSE OUTCOMES:

On completion of the course students will be able to

1. Design, develop, and implement a compiler for any language.
2. Use Lex and Yacc tools for developing a scanner and a parser.
3. Design and implement LL and LR parsers.
4. Develop a type of checker for a given language.
5. Implement storage allocation technique for a given program.

CO-PO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	2	3					2		1
CO2		2	2	2	3							1
CO3		2	2	2	3							1
CO4		2	2	2	3							1
CO5	1	2	2	2	3							1

****END****